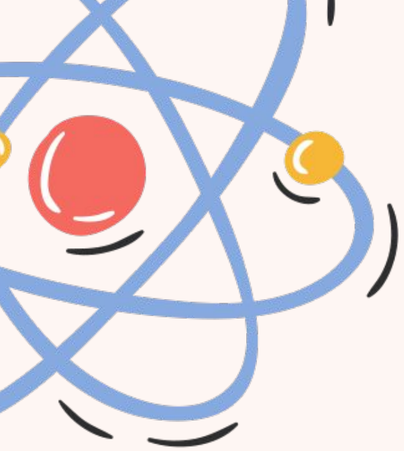
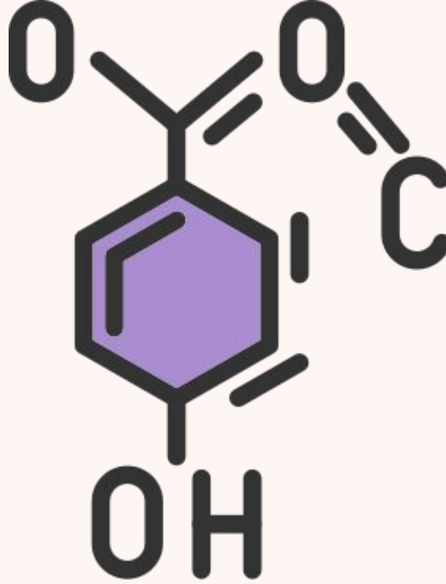


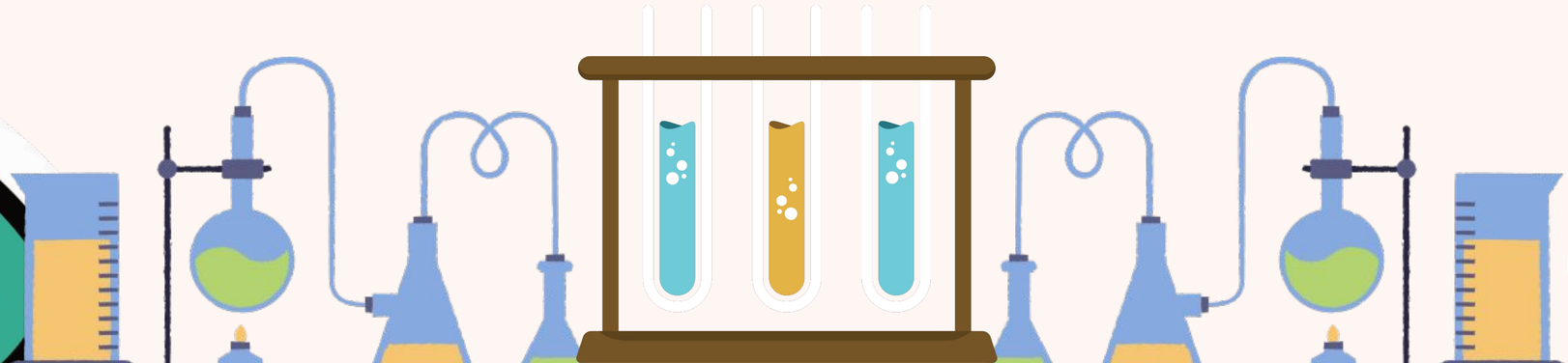


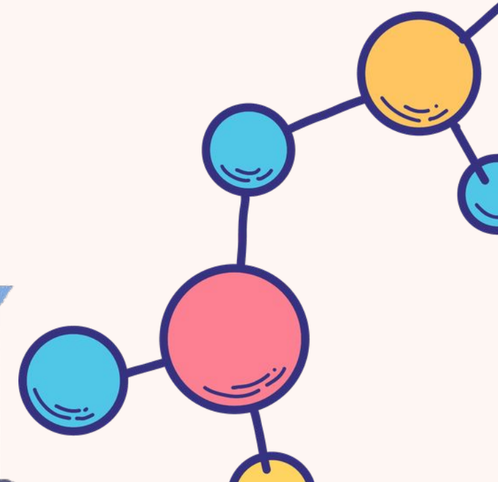
NL01CHEM01:

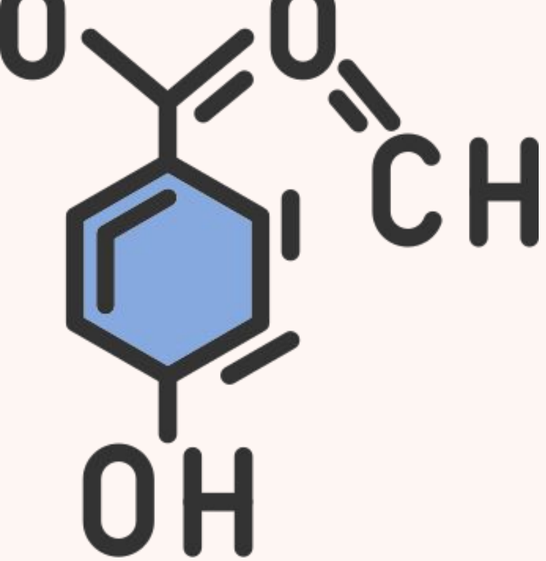
Introduction to Chemistry for Middle School Students Session 1

Nanum Learn



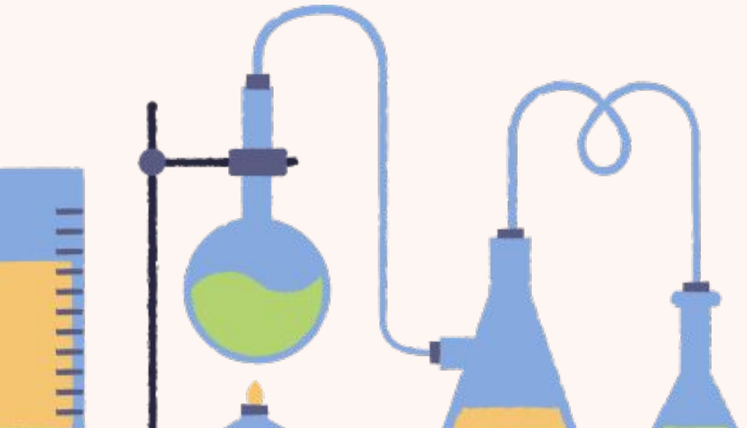


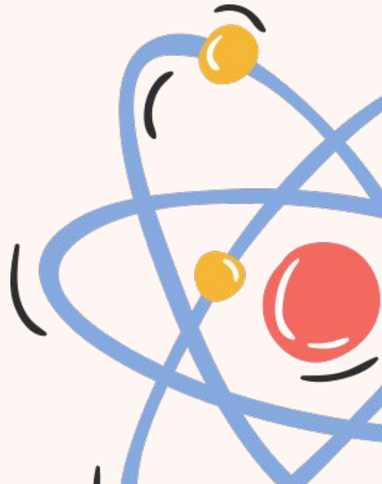


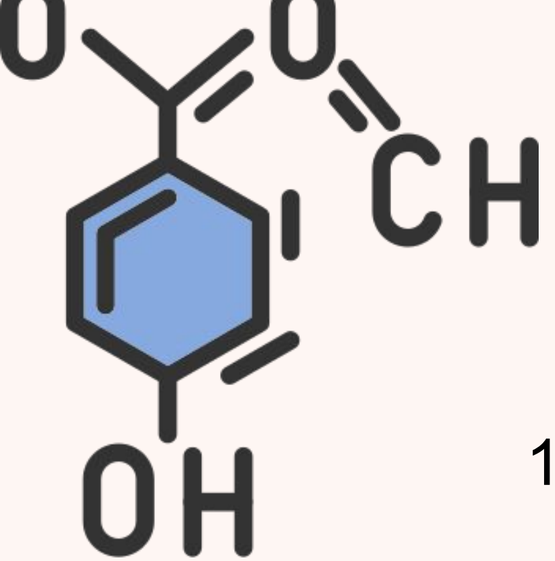


Ice Breaker

If you are safe to eat any chemical elements, what would you try?



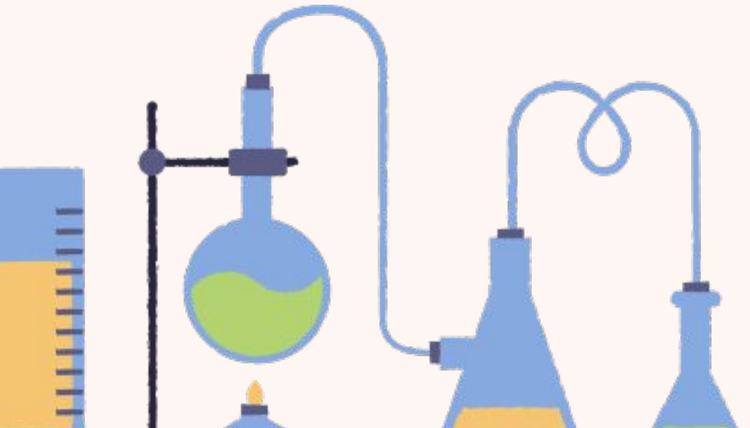


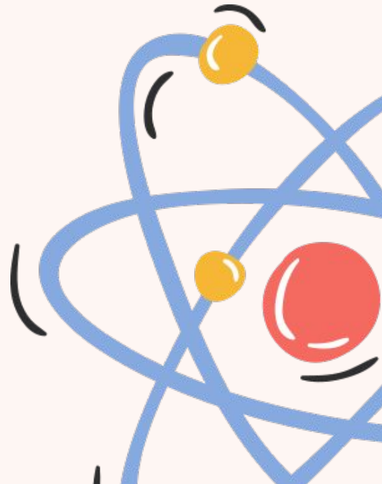


Agenda



1. About Nanum Learn
2. Chemistry Nanum
3. Breakout Rooms





Our Leadership



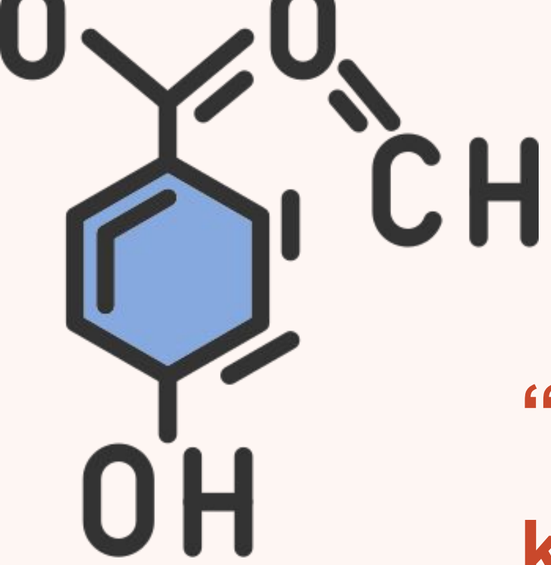
Enoch Yu
Founder & President



William Kang
Chemistry Director



Jian Choi
Communication Officer

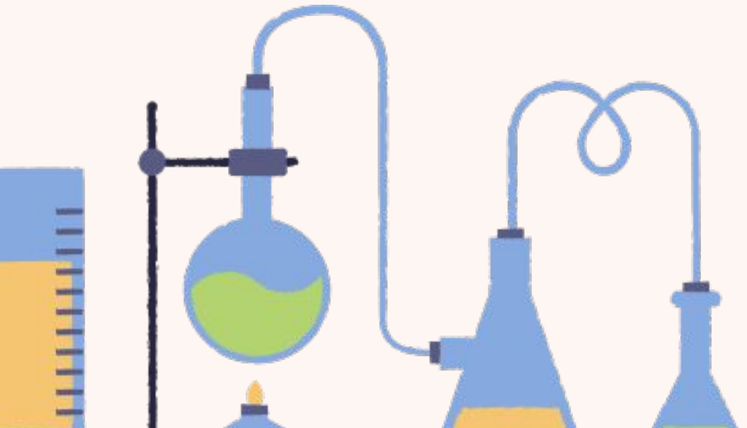


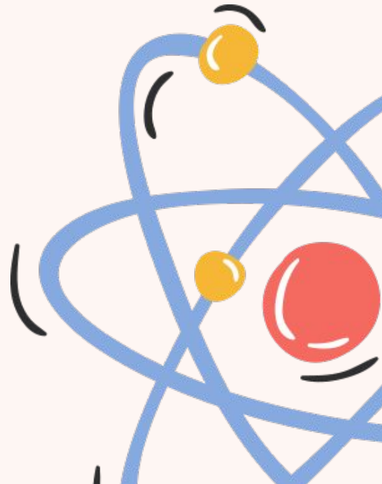


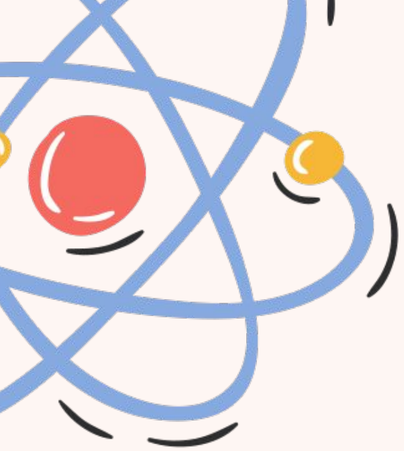
What we do

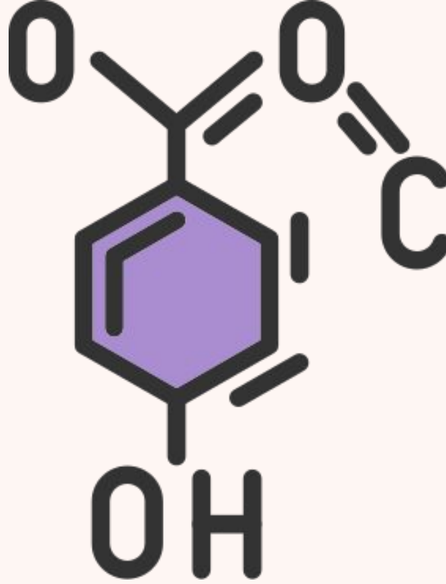
“To inspire global learners by returning knowledge and skills we have received”

1. Learn & Imagine
2. Create (Resources, Books)
3. Share (**Nanum**, Handouts)





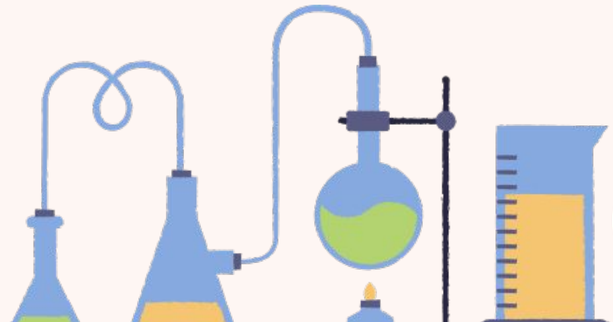


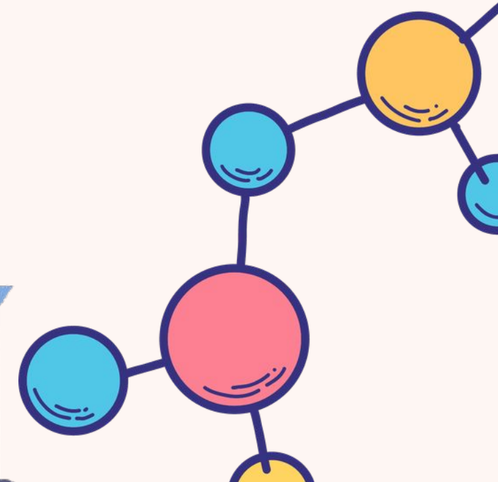


Introduction to Chemistry for Middle School Students Session 1







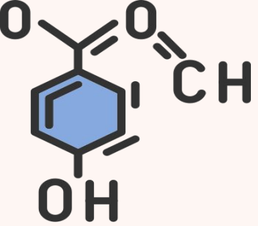


How can a measured property help us identify an unknown material?

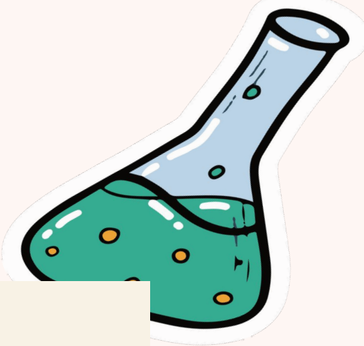


By end of Nanum today, we will be able to:

- Classify matter using composition and particle diagrams
- Distinguish physical change from chemical change by considering substance identity
- Calculate density, mass, or volume and justify the units



Three composition ways



Element

One kind of atom — alone or bonded to itself.



Photo by news central asia
https://www.newscentralasia.net/2026/07/27/critical-minerals-and-rare-earth-elements-understanding-the-resources-shaping-the-future/

Example

Oxygen gas — both atoms in O_2 are oxygen.

Compound

Two or more kinds, in a fixed ratio, bonded into one substance.

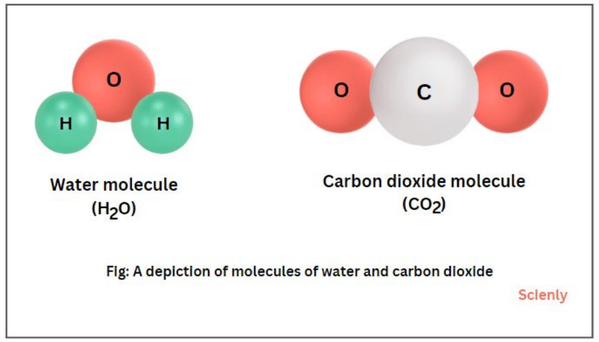


Photo by Sciencly
https://scienly.com/what-is-compound/

Example

Oxygen gas — both atoms in O_2 are oxygen.

Mixture

More than one substance together, with a ratio that can vary.



Photo by LearnsWithSweta
https://www.instagram.com/p/DN9rxEeD_N2/

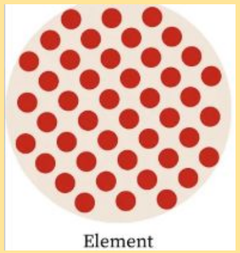
Example

Air and salt solution

Uniform isn't the same as pure.

Pure water

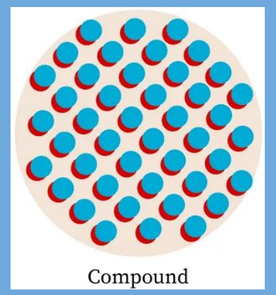
One substance.



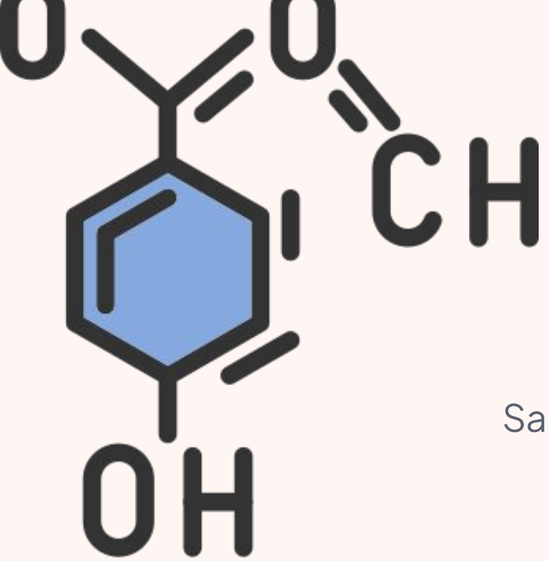
Same kind of molecule — water, water, and once more, water.

Saltwater — uniform, not pure

Two substances — just well-mixed.



Looks clear. Holds water — and salt.



Change the arrangement — or change the substance?



Same atoms can be rearranged without becoming anything new.

Physical change

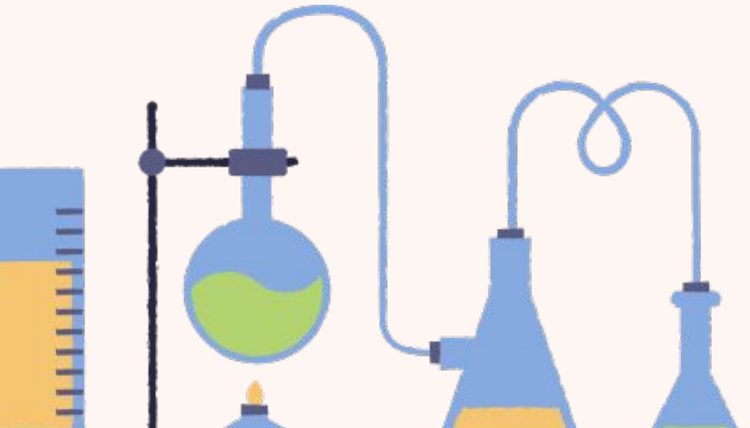
Example: ice → water.
Still H_2O.

Other

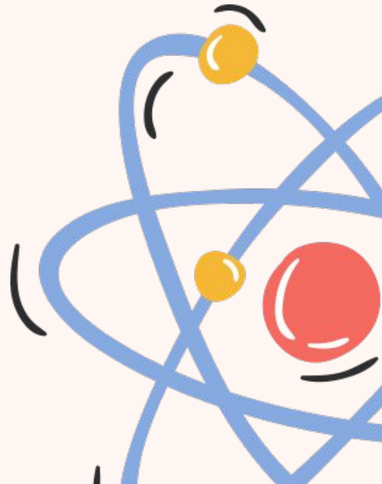
Chemical change

Example: iron + oxygen
→ rust.

Different substance — new identity.



Bubbling alone isn't enough. Boiling water bubbles too — and it is still water.



Three numbers that name what fits in your hand

Same atoms can be rearranged without becoming anything new.



Mass

How much stuff.

g

Read on a balance.
Grams — and kilograms when useful.

Volume

How much room.

mL

= cm^3

One mL fills the same space
as one cubic centimetre.

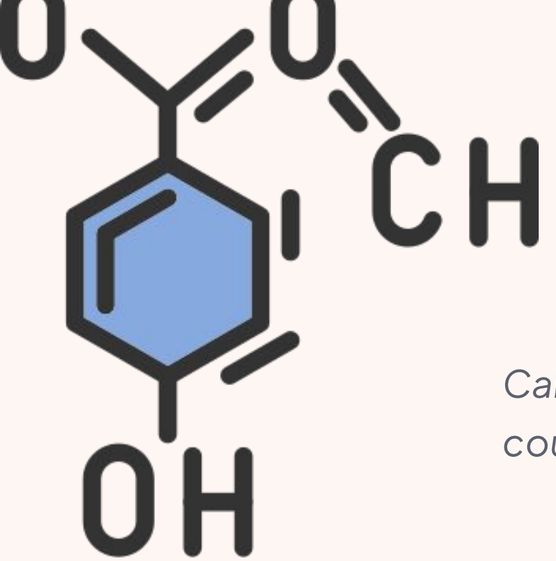
Density

How much stuff in a fixed space.

$\rho = m / V$

Same material under the same conditions —
double the sample, you double the mass
and the volume, and the density stays.

grams per cubic centimeter



Three samples on the bench.



Calculate each density. Then ask the question — could any of these be the same material?

Sample A

A

mass **24 g**
volume **8 cm**

24 g ÷ 8 cm³

3.0 g/cm³

Sample B

B

mass **45 g**
volume **15 cm**

45 g ÷ 15 cm³

3.0 g/cm³

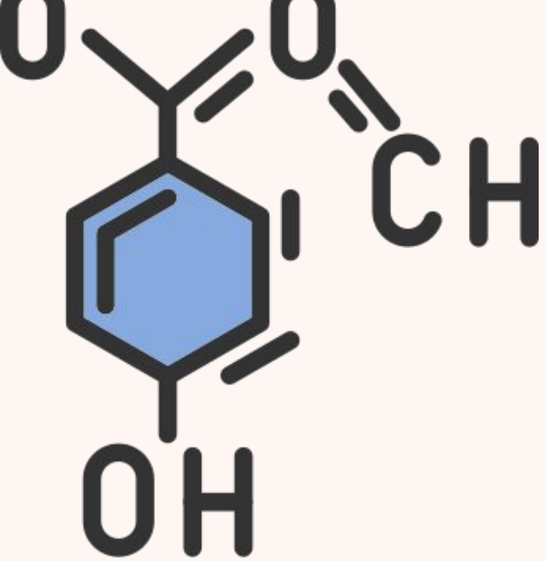
Sample C

C

mass **24 g**
volume **12 cm³**

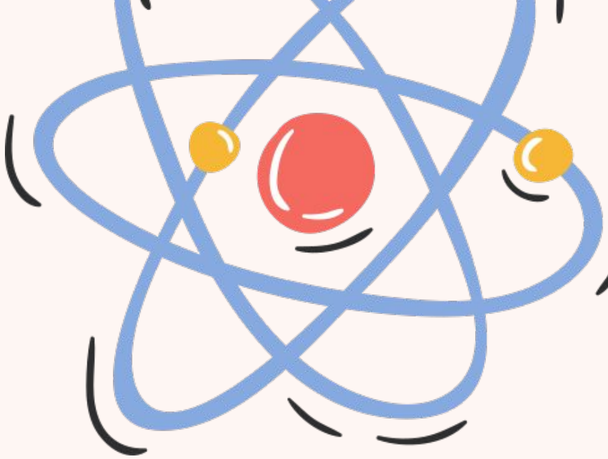
24 g ÷ 12 cm³

2.0 g/cm³

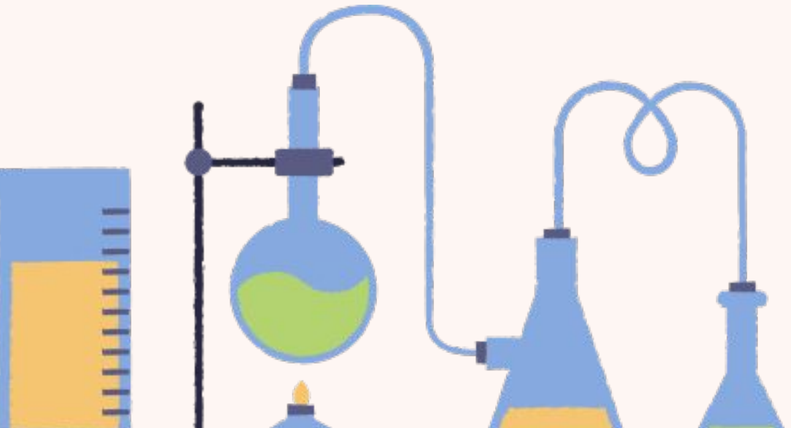


Atomic Structure.

— Isotopes & Ions



Which changes affect an atom's identity, mass number, or charge?

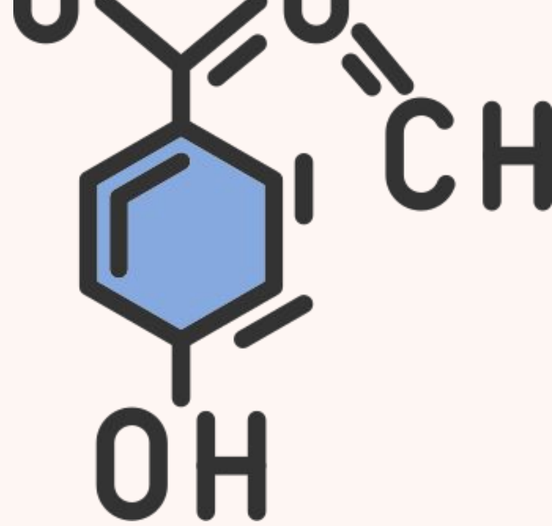




Inside the atom

Three particles — and where they live.

A simple model: protons and neutrons in the nucleus, electrons spread around them.



One

Proton

In the nucleus.

+1

`charge +1`
`mass ~1 amu`

Two

Neutron

In the nucleus too.

0

`charge 0`
`mass ~1 amu`

Three

Electron

Outside the nucleus.

−1

`charge −1`
`mass ≈ 1/1836 amu`

A neutral atom has equal proton and electron counts.

The proton count chooses the element.

A whole-number sum: protons plus neutrons.

Mass number is the count you can do with fingers.
The decimal mass on the periodic table is something else.

$$A = p + n$$

So, $n = A - Z$.

Subtract the atomic number from the mass number to find the neutron count.

Z=p

A worked example

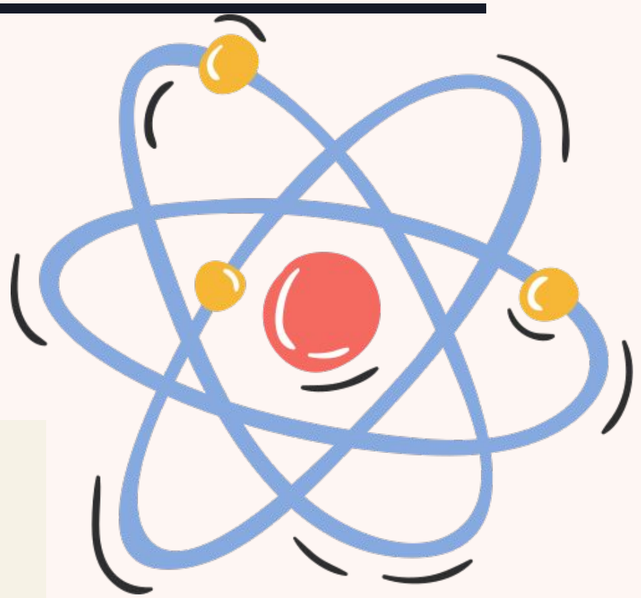
Carbon-12

atomic number Z = **6**

mass number A = **12**

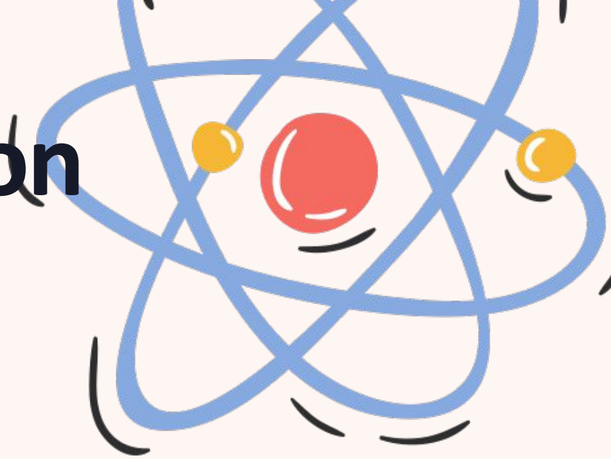
neutrons n = **A – Z = 6**

Six protons, six neutrons. Mass number 12.



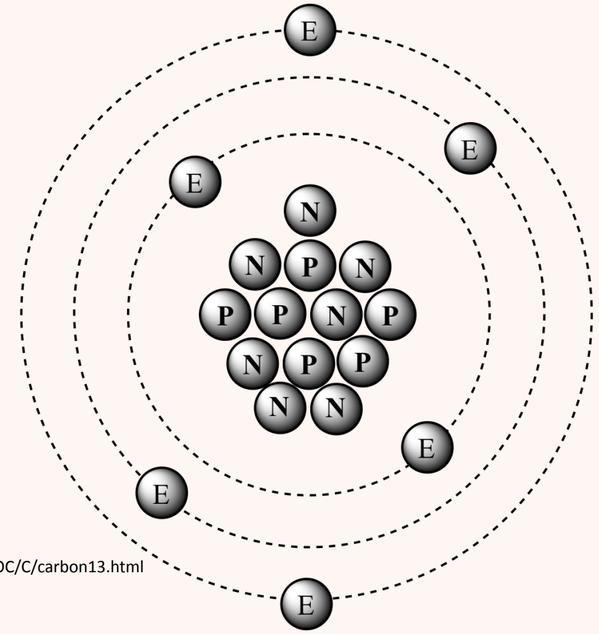


Same protons — different neutron counts.



Carbon-12

protons **6**
neutrons **6**
mass **12**



Same element, heavier

Carbon-14

6 protons
6 neutrons
14 mass

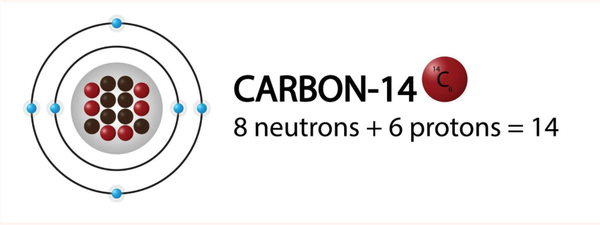
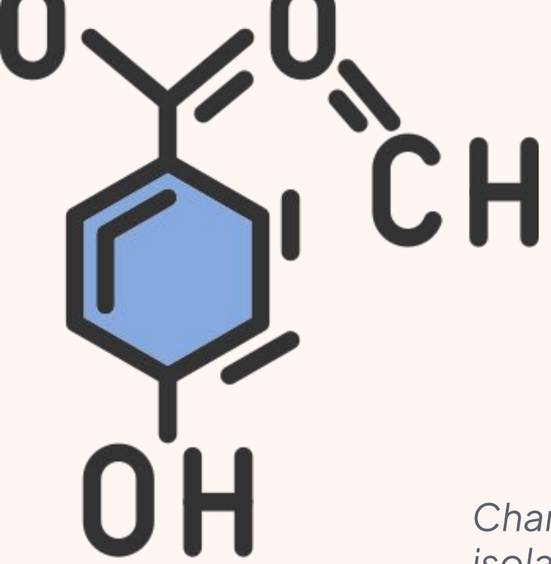
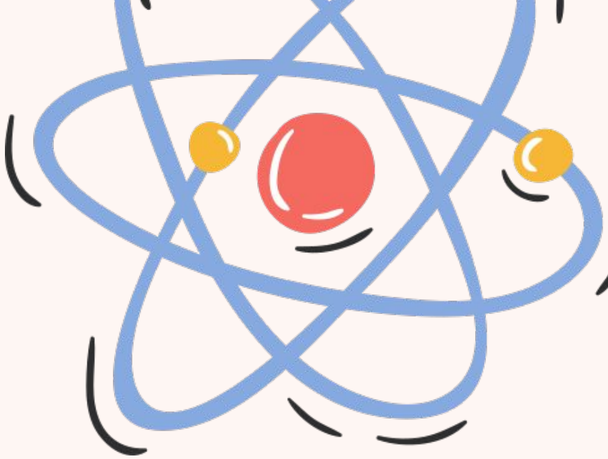


Photo by UCLA

Source: https://www.chem.ucla.edu/~harding/IGOC/C/carbon13.html



Charge shows up when the electron count shifts.



Charge equals the difference between proton and electron counts — not a property memorised in isolation.

$$q = p - e$$

Lose an electron — go positive

Na

11 p + 11 e
charge **0**

11 p + 10 e
charge **+1**

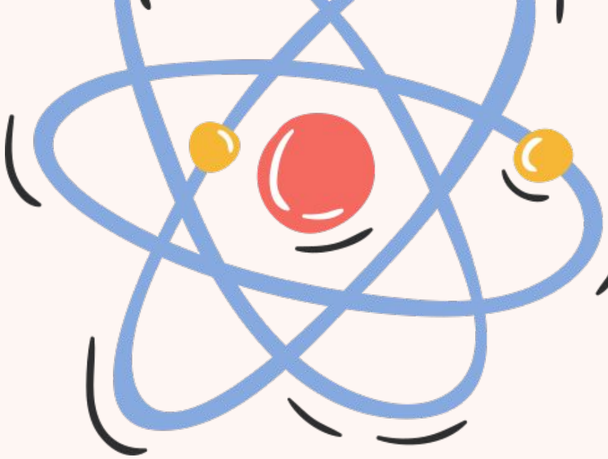
Gain an electron — go negative

Cl → Cl⁻

17 p + 17 e
charge **0**

17 p + 18 e
charge **−1**

Count, then explain the comparison.

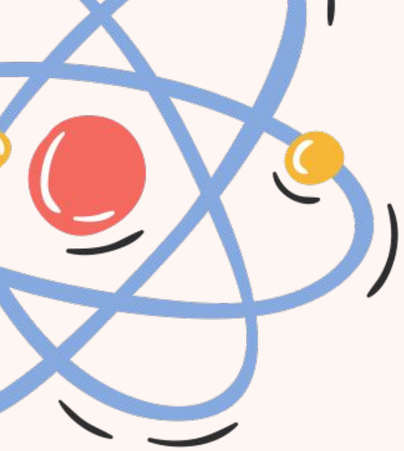


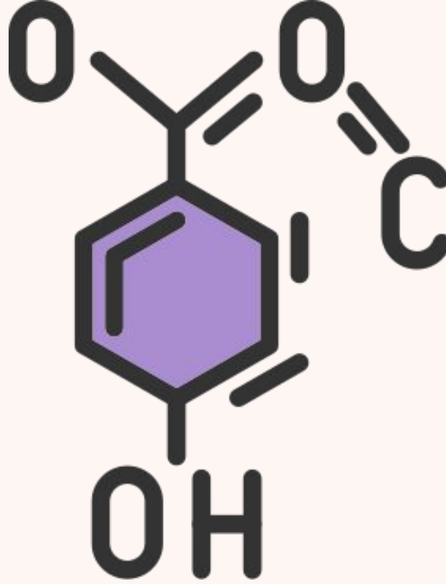
ENTITY	P	N	E	CHARGE
Carbon-12	6	6	6	0
Carbon-14	6	8	6	0
Sodium ion Na^+	11	12	10	+1
Chloride Cl^-	17	18	18	−1

ISOTOPES
CHANGE N

IONS
CHANGE E

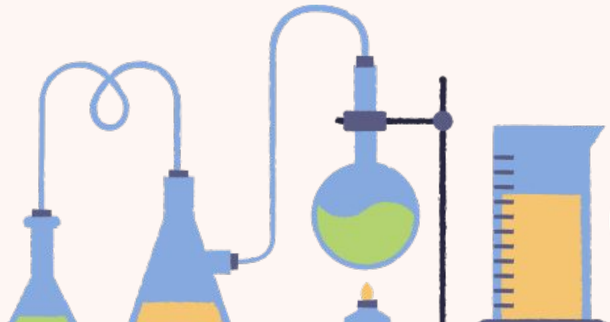
Breakout Rooms

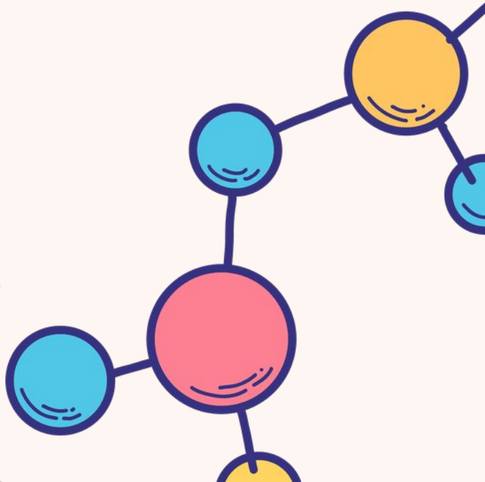


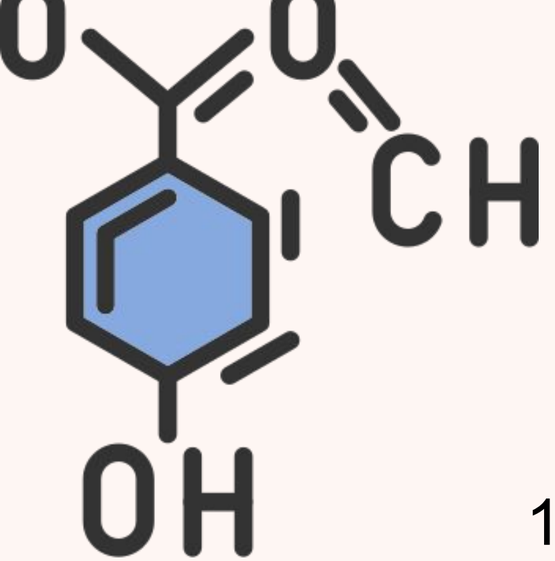








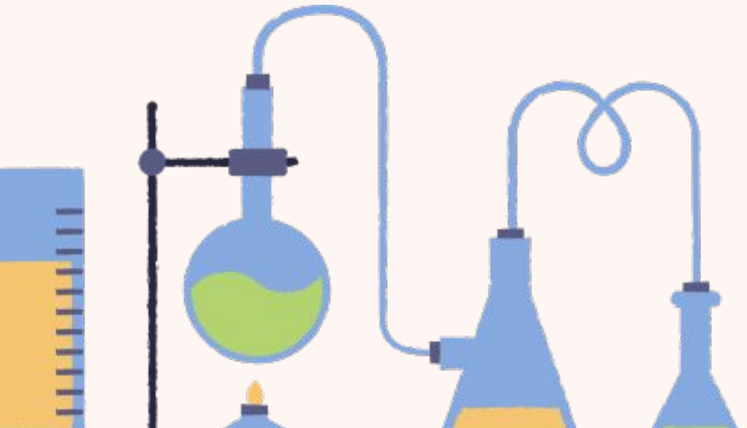


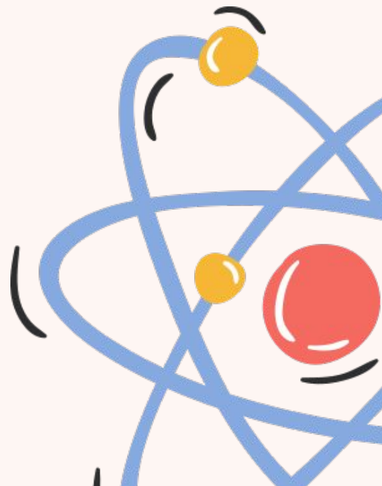


Wrap-Up



1. Subscribe to Nanum Learn
2. Check Website





See You Next Tuesday!

